

Sales Prediction using Machine Learning

Project Objective

- Build a model to predict sales.
- Analyze business factors affecting sales.
- Learn forecasting using regression.
- Improve predictive analytics skills.

Dataset

- Source: Kaggle
- Format: CSV
- Contains Product, Sales, Advertising, Region, Marketing and other business features.

Programming Language

- Python

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Workflow

- Import libraries.
- Load dataset.
- Inspect data.
- Handle missing values.
- Remove duplicates.
- Handle outliers.
- Perform EDA.
- Create visualizations.
- Select features.
- Train/test split.
- Train regression model.
- Predict sales.
- Evaluate with MAE, MSE, RMSE and R^2 .

Data Cleaning

- Removed duplicates.
- Filled missing values.
- Processed categorical variables.
- Verified data quality.

Visualizations

- Histograms
- Bar Charts
- Scatter Plots
- Heatmaps
- Box Plots

Skills Learned

- Business Analytics
- Data Cleaning
- Regression
- Predictive Analytics
- Model Evaluation

- Python Programming

Applications

- Inventory Planning
- Demand Forecasting
- Marketing Strategy
- Revenue Forecasting

Future Enhancements

- Time-series models
- Ensemble learning
- Dashboard deployment